

ADAPTIVE CARE

Product design system

V2.1 · Warm Indigo

A production rulebook for a calm, precise care-operations product. Built to make resident risk, device trouble, uncertainty, ownership, and the next staff action immediately clear.

							
Warm canvas	#F7F6F2	Working white	#FFFFFF	Deep ink	#1F2328	Warm indigo	#4D55C5
							
Positive	#176B57	Watch	#8A4D0F	Resident risk	#B4232D	Device and data	#315E91

Quiet at rest. Unmistakable when action is required.

Adaptive Care design system

Version: 2.1 **Direction:** Warm Indigo **Status:** Approved foundation for new frontend work **Primary mode:** Clinic operations **Secondary mode:** Home and family experience

This document is the machine-readable source of truth for Adaptive Care UI work. It replaces every earlier color, type, spacing, radius, shadow, and component rule.

The system must help a staff member answer one question within two seconds:

Which resident needs my attention right now?

The interface is calm because the work is stressful. It is dense because staff scan many residents. It is precise because missing or unreliable information can change a care decision.

1. Product character

Adaptive Care should feel calm, warm, technical, dense, and honest. It must never feel sleepy, decorative, cryptic, cramped, or falsely certain.

The visual identity uses warm neutral backgrounds, white working surfaces, deep indigo actions, compact typography, hairline structure, and semantic colors reserved for meaning.

Do not imitate a hospital alarm console. Do not imitate a generic pastel SaaS dashboard. Do not create a dark command center.

2. Product modes

Clinic operations

Clinic screens optimize for repeated scanning, fast triage, keyboard use, clear ownership, and reliable state changes. Lists and tables are the main layout pattern.

Home and family

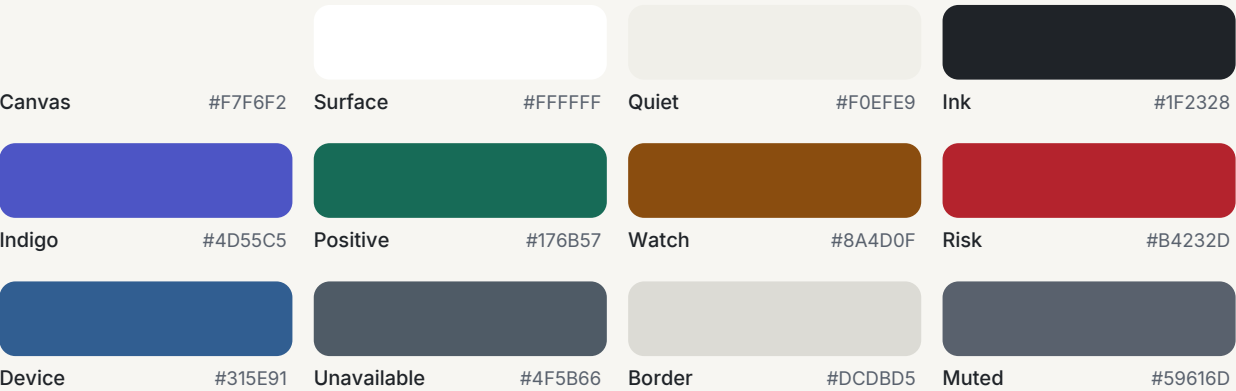
The home product inherits the font, color roles, controls, focus behavior, and accessibility rules. It uses more space, fewer simultaneous facts, simpler language, and no clinic queue density.

Sharing the design system does not mean sharing page composition.

3. Non-negotiable rules

- 1 Resident risk, device health, data confidence, monitoring condition, and workflow state are separate facts.
- 2 Red is reserved for resident-risk escalation and destructive validation. Device failure does not become red unless policy separately marks its operational priority as high or critical.
- 3 Missing, stale, limited, or unavailable data never looks normal.
- 4 Color never carries meaning alone. Use a label and a stable icon or structural treatment.
- 5 The attention queue leads the clinic overview. Supporting metrics never compete with it.
- 6 Repeated records use lists or tables. Cards hold summaries, focused tasks, or progressive detail.
- 7 One decision area has one primary action and no more than two visible secondary actions.
- 8 Components use semantic tokens. Page CSS does not use raw hex values.
- 9 Every interactive component has keyboard, focus, loading, disabled, error, and responsive behavior.
- 10 The real rendered screen must pass visual inspection before it is complete.

4. Color architecture



Color has three layers:

- 1 **Foundation values** are the raw scales.
- 2 **Role tokens** describe interface jobs such as canvas, text, border, and action.
- 3 **Domain tokens** describe product meaning such as resident risk, device health, confidence, and workflow.

Components consume role and domain tokens. They do not consume foundation values directly.

4.1 Warm neutral foundation

Token	Value	Use
warm-0	#FFFFFF	Working surface
warm-25	#FCFCFA	Raised neutral surface
warm-50	#F7F6F2	Main canvas
warm-100	#F0EFE9	Sidebar and quiet fill
warm-150	#E9E8E2	Hover fill
warm-200	#DCDBD5	Hairline structure
warm-300	#C8C8C1	Strong border
warm-400	#A0A19B	Disabled decoration
warm-500	#73767A	Tertiary text, only at 14px or larger
warm-600	#59616D	Secondary text
warm-700	#414851	Strong secondary text
warm-800	#2C3239	Headings

Token	Value	Use
warm-900	#1F2328	Primary text
warm-950	#14171B	Highest contrast ink

4.2 Brand indigo foundation

Token	Value	Use
indigo-25	#FBFBFF	Faint wash
indigo-50	#F4F4FF	Selected background
indigo-100	#EAEBFF	Strong selected background
indigo-200	#D7D9FF	Selected border
indigo-300	#B5B9F4	Decorative data accent
indigo-400	#858BE3	Non-text accent
indigo-500	#636AD4	Brand mark
indigo-600	#4D55C5	Primary action and focus
indigo-700	#3F46A8	Hover and accessible brand text
indigo-800	#333888	Pressed action
indigo-900	#292D6A	Strong brand ink

Indigo means navigation, selection, focus, or an available staff action. It never means severity.

4.3 Semantic families

Every semantic family contains a soft background, border, text and icon color, and strong color.

Family	Soft	Border	Text and icon	Strong
Positive	#E7F5F0	#A9D8C8	#176B57	#0F5746
Watch	#FFF2DD	#E8C58D	#8A4D0F	#6D3908
Resident risk	#FDEBED	#E4A0A7	#B4232D	#8F1721
Device and data	#EAF2FB	#AEC9E5	#315E91	#244A73
Unavailable	#EEF1F2	#C8D0D3	#4F5B66	#39444D

Approved small-text pairs exceed WCAG 2.2 AA contrast:

- positive text on positive soft: 5.71:1;
- watch text on watch soft: 6.03:1;
- resident-risk text on resident-risk soft: 5.68:1;
- device text on device soft: 5.92:1;
- unavailable text on unavailable soft: 6.12:1;
- white text on primary indigo: 6.14:1.

4.4 Role tokens

```
:root {
  color-scheme: light;
  --ac-canvas: #F7F6F2;
  --ac-surface: #FFFFFF;
  --ac-surface-raised: #FCFCFA;
  --ac-surface-quiet: #F0EFE9;
  --ac-surface-hover: #E9E8E2;
  --ac-text-primary: #1F2328;
  --ac-text-secondary: #59616D;
  --ac-text-strong-secondary: #414851;
  --ac-text-disabled: #73767A;
  --ac-text-inverse: #FFFFFF;
  --ac-border-subtle: #DCDBD5;
  --ac-border-strong: #C8C8C1;
  --ac-border-interactive: #A0A19B;
  --ac-action: #4D55C5;
  --ac-action-hover: #3F46A8;
  --ac-action-pressed: #333888;
  --ac-action-soft: #F4F4FF;
  --ac-action-soft-strong: #EAEBFF;
  --ac-action-border: #D7D9FF;
  --ac-focus: #4D55C5;
  --ac-positive-bg: #E7F5F0;
  --ac-positive-border: #A9D8C8;
  --ac-positive-text: #176B57;
  --ac-positive-strong: #0F5746;
  --ac-watch-bg: #FFF2DD;
  --ac-watch-border: #E8C58D;

  --ac-watch-text: #8A4D0F;
  --ac-watch-strong: #6D3908;
  --ac-risk-bg: #FDEBED;
  --ac-risk-border: #E4A0A7;
  --ac-risk-text: #B4232D;
  --ac-risk-strong: #8F1721;
  --ac-device-bg: #EAF2FB;
  --ac-device-border: #AEC9E5;
  --ac-device-text: #315E91;
  --ac-device-strong: #244A73;
  --ac-unavailable-bg: #EEF1F2;
  --ac-unavailable-border: #C8D0D3;
  --ac-unavailable-text: #4F5B66;
  --ac-unavailable-strong: #39444D;
}
```

4.5 Color rules

- Normal monitoring uses primary text plus a small positive icon or word. Do not tint the whole row green.
- Critical resident risk may use a solid red action or a soft red row treatment. High risk uses the soft treatment. Both keep explicit labels.
- Device-only problems use the device family and a hardware icon. A separate priority field may elevate the item.

- Low confidence uses the device and data family with an explicit reason.
- Unavailable uses the unavailable family. Never render last-known values as current.
- Decorative supporting cards may use indigo washes. Do not use semantic colors as decoration.
- Dark mode is out of scope for V2.1. Forced-colors and high-contrast modes are required.

5. Typography

PAGE TITLE 28/34

Clinic overview

MAJOR HEADING 22/28

Residents needing attention

RECORD TITLE 15/21

Resident B · Room 102

BODY 14/20

Monitoring is active. Review the newest event and data quality.

METADATA 12/17

Updated 8:31 PM · 8 minutes ago · Assigned to Maya Chen
Use **Inter Variable** for the full product. Its text optical size supports small interface text, its tall x-height aids scanning, and its tabular numbers align times, counts, durations, and measurements.
Self-host WOFF2 files. Do not depend on a third-party font request at runtime.

```
html {
  font-family: "Inter Variable", Inter, ui-sans-serif, system-ui, -apple-system,
    BlinkMacSystemFont, "Segoe UI", sans-serif;
  font-optical-sizing: auto;
  font-synthesis: none;
  text-rendering: optimizeLegibility;
}
.numeric { font-variant-numeric: tabular-nums lining-nums; }
.identifier { font-variant-numeric: tabular-nums slashed-zero; }
```

Type scale

Role	Size / line height	Weight	Use
Display	32 / 38px	650	Product or empty-state moments only
Page title	28 / 34px	650	One per page
Major heading	22 / 28px	650	Attention or detail section
Section heading	18 / 24px	650	Panel and table groups
Record title	15 / 21px	600	Resident, event, device
Body	14 / 20px	400	Default reading text
Body strong	14 / 20px	550	Important explanation

Role	Size / line height	Weight	Use
Label	13 / 18px	550	Controls and table headings
Metadata	12 / 17px	450	Time, room, source, secondary facts
Overline	11 / 16px	650	Short section label only

Do not use meaningful text smaller than 12px. Use sentence case. Long safety text wraps and remains available at 200% zoom.

6. Spacing and density

Use a 4px base grid.

```
:root {  
  --ac-space-0: 0;  
  --ac-space-0-5: 2px;  
  --ac-space-1: 4px;  
  --ac-space-1-5: 6px;  
  --ac-space-2: 8px;  
  --ac-space-3: 12px;  
  --ac-space-4: 16px;  
  --ac-space-5: 20px;  
  --ac-space-6: 24px;  
  --ac-space-8: 32px;  
  --ac-space-10: 40px;  
  --ac-space-12: 48px;  
  --ac-space-16: 64px;  
  --ac-space-20: 80px;  
}
```

- **Comfortable** is the default. Controls are 40px high and rows are 60 to 68px.
- **Compact** is an opt-in desktop preference. Controls are 36px high and rows are 48 to 56px. Text and semantic labels do not shrink.
- **Touch** activates for coarse pointers or narrow screens. Primary targets are at least 44 by 44px.

7. Geometry, borders, elevation, and motion

```
:root {
  --ac-radius-xs: 4px;
  --ac-radius-sm: 6px;
  --ac-radius-control: 8px;
  --ac-radius-card: 10px;
  --ac-radius-popover: 12px;
  --ac-radius-dialog: 14px;
  --ac-radius-pill: 999px;
  --ac-shadow-card: 0 1px 2px rgb(20 23 27 / 4%);
  --ac-shadow-popover: 0 8px 24px rgb(20 23 27 / 12%), 0 2px 6px rgb(20 23 27 / 8%);
  --ac-shadow-dialog: 0 24px 64px rgb(20 23 27 / 18%), 0 6px 18px rgb(20 23 27 / 10%);
  --ac-overlay: rgb(20 23 27 / 42%);
  --ac-duration-instant: 80ms;
  --ac-duration-fast: 120ms;
  --ac-duration-standard: 160ms;
  --ac-duration-emphasis: 240ms;
  --ac-ease-standard: cubic-bezier(.2, 0, 0, 1);
  --ac-ease-exit: cubic-bezier(.4, 0, 1, 1);
}
```

Base surfaces use borders, not shadows. Shadows identify overlap. Motion explains focus, selection, loading, and saved state. Do not pulse critical rows.

8. Layout system

Desktop shell

- Sidebar: 232px.
- Top bar: 56px.
- Page max width: 1600px.
- Page padding: 32px.
- Main content gap: 24px.
- Detail split: `minmax(0, 1fr) 360px`.

Breakpoints

Width	Behavior
1440px and wider	Full table, context rail, 32px page padding
1200 to 1439px	Full table with lower-priority columns compressed
960 to 1199px	Collapsible sidebar, detail rail moves below or into sheet
768 to 959px	Navigation drawer, simplified table, 24px padding
Below 768px	Ordered record list, full-screen filters and details, 16px padding

Column survival order:

- 1 Resident and room
- 2 Attention reason
- 3 Resident risk
- 4 Confidence and freshness
- 5 Workflow state and owner
- 6 Primary action
- 7 Device detail
- 8 Secondary timestamps and metadata

Never hide the first six without placing the information inside the same record.

9. State model

Do not create one generic **status** property.

Axis	Values
Resident risk	critical, high, watch, none
Monitoring condition	active, away, paused, possible_multi_person, calibrating, limited, unavailable
Data confidence	high, medium, low, unavailable
Device health	online, degraded, buffering, retrying, offline, assignment_unavailable, not_yet_available
Workflow	new, acknowledged, assigned, checked, resolved, overdue
Freshness	current, delayed, stale, unknown

Confidence includes a reason when not high. Freshness shows relative age and makes exact time available.

Precedence:

- 1 Critical and high resident risk lead the hierarchy.
- 2 Unavailable or stale evidence stays visible beside risk.
- 3 Device failure never erases a resident event.
- 4 Workflow ownership does not lower resident risk.
- 5 Resolved events leave the active queue and remain immutable.
- 6 A recurrence creates a new linked event.

10. Attention item

Attention queue			4 active items · Updated now
Resident B Room 102 · Unusual movement	Critical	8 min · Unassigned	Review
Resident E Room 105 · Device offline	Device	17 min · Facilities	Review
Resident D Room 104 · Data limited	Watch	4 min · Maya Chen	Review
Resident A Room 101 · Monitoring active	Active	Current	Review

The attention item is the main clinic component. It contains resident and room, direct reason, resident risk, confidence and explanation, freshness, device condition when relevant, workflow and owner, elapsed time, one primary next action, and overflow utilities.

Default sorting:

- 1 Critical risk
- 2 High risk
- 3 Overdue work
- 4 Oldest unacknowledged item
- 5 Watch items
- 6 Device-only operational items

Background refresh must not move the row under the pointer or keyboard focus. Announce meaningful queue changes in a polite live region.

11. Application shell

- Navigation stays quiet and uses icon plus text.
- Keep clinic navigation to six or fewer top-level destinations.
- Active navigation uses indigo text, icon, and a soft indigo background.
- Keep synthetic-data labeling visible in demo builds.
- Use one page-width banner for global outages.
- Do not repeat an alert in the top bar, banner, and page content.

12. Buttons

Primary uses deep indigo with white text. Secondary uses a white surface and strong border. Ghost is transparent until hover. Destructive uses soft red and risk text; solid red is reserved for confirmed destructive or critical workflow action.

Required states are default, hover, pressed, focus-visible, loading, disabled, success, and error. Loading retains label and width. Disabled actions explain why when the reason is not obvious.

13. Inputs and forms

- Labels remain visible. Placeholder text is an example, never the label.
- Default controls are 40px high. Text areas show at least three lines.
- State required and optional fields in words.
- Place validation beside the field and add an error summary for long forms.
- Failed submissions preserve valid input.
- Disabled and read-only look and behave differently.
- Use native semantics before custom ARIA.

Required components are text field, search, textarea, select, combobox, checkbox, radio group, switch, date/time input, and segmented control.

14. Tables and operational lists

- Use native table structure on wide layouts and an accessible caption.
- Sort buttons live inside headers and announce direction.
- Tab reaches sortable headers, primary links, checkboxes, and actions, not inert cells.
- Search and common filters share one toolbar with no more than five exposed actions.
- Active filters appear as removable chips with Clear all.
- Empty, filtered-empty, loading, error, and stale are different states.
- Missing values say Not measured, Not applicable, Unavailable, or Stale.
- Never show blank or zero as a substitute for missing data.
- Do not make a whole row clickable when it contains nested actions.

15. Panels, sheets, dialogs, and popovers

Use a non-modal desktop split panel for quick inspection when the queue must remain visible. On narrow screens use a modal sheet or full-screen dialog with focus trap, Escape support, labelled title and description, and focus return.

Detail order is identity, attention summary, risk and confidence, freshness and device health, evidence, workflow history, and actions.

Use popovers only for light filters and utility previews. Consequential actions use dialogs.

16. Feedback and system messages

Scope	Pattern
Whole service	One page-width banner
One panel or section	Inline message
One field	Field error and description
Short saved confirmation	Toast plus live-region announcement
Consequential decision	Dialog

Do not use a toast as the only record of a critical failure. Do not duplicate a local validation error in a page banner.

17. Loading, empty, error, and stale

- Initial loading uses a skeleton with stable headers and approximate rows.
- Genuine empty explains what the absence means.
- Filtered empty offers Clear filters.
- Error keeps useful context, names the problem, and offers Retry.
- Stale keeps the last successful values visible, labels their age and cause, and qualifies actions that need current evidence.

Never replace stale data with zero, normal, or a fresh-looking skeleton.

18. Event workflow

- **Acknowledge** records actor and time without lowering risk.
- **Assign** shows the responsible staff member and preserves reassignment history.
- **Check** records that the resident was checked without automatically resolving.
- **Resolve** requires an outcome and records actor and time. High and critical resolution requires confirmation. Resolved records are read-only.
- **Feedback** separates staff observation from AI interpretation and sensor evidence.

Every save supports pending feedback, duplicate-action protection, failure recovery, and server-conflict refresh.

19. Accessibility contract

WCAG 2.2 AA is the release floor.

- Normal text is at least 4.5:1.
- Large text is at least 3:1.
- Required control boundaries, icons, and states are at least 3:1.
- Focus uses a 2px solid indigo ring with a 2px offset and is never obscured.
- Primary touch targets are at least 44 by 44px.
- Dense desktop controls never fall below 24px and maintain sufficient spacing.
- Status uses words plus icons or structure.
- Screen readers receive meaningful changes, not every polling refresh.
- The main flow works by keyboard.
- At 200% zoom, risk, reason, freshness, owner, and action remain present.
- At 400% zoom, content reflows except genuine data tables.
- Reduced motion removes transforms and workflow movement.
- Forced-colors mode preserves borders, focus, icons, and labels.

20. Icons

Use one outlined icon family with 1.75 to 2px strokes and sizes of 16, 18, 20, or 24px. Icon-only buttons require names and tooltips.

Semantic roles:

- critical: alert octagon;
- high: alert triangle;
- watch: eye or clock;
- device issue: signal off or hardware icon;
- low confidence: question in circle;
- active or resolved: check circle;
- stale: clock with an explicit Stale label.

Do not use medical symbols to imply diagnosis.

21. Data visualization

Use a chart only when it answers a comparison or trend question faster than text. "4 of 6 online" is better than a donut.

Every chart includes a plain-language summary and accessible data table. Missing and stale data are explicit. Tooltips are keyboard reachable. Charts remain clear in grayscale and print.

22. Content rules

Write direct operational language.

Good:

- "Resident B needs review."
- "Monitoring unavailable. Room device has been offline for 12 minutes."
- "Low confidence because room occupancy is unclear."
- "Saved. Event assigned to Maya Chen."

Avoid false reassurance, unsupported diagnoses, vague errors, and unexplained technical words.

Label AI interpretation as interpretation. Keep sensor evidence, deterministic warnings, AI text, and staff notes visually separate.

23. Page recipes

- **Clinic overview:** compact header, attention queue, resident inventory, then coverage and device context.
- **Residents:** header, search and filters, operational table, detail panel or route.
- **Resident detail:** identity, current monitoring truth, attention and actions, evidence, device context, history, settings.
- **Events:** header, queue controls, priority list, detail, workflow actions, immutable history.
- **Devices:** fleet summary, device list, health, assignment, last contact, limitations, recovery history.
- **Scenario lab:** one ordered scenario list and one result panel with synthetic provenance.
- **Home overview:** one plain-language answer, latest meaningful update, simple trends, and routine context.

24. Component ownership

Build shared Adaptive Care components around accessible behavior primitives. Radix Primitives is the preferred base for dialog, popover, tooltip, select, tabs, and menu behavior. It does not provide product styling or meaning.

Each shared component documents purpose, anatomy, variants, state matrix, token mapping, keyboard behavior, screen-reader behavior, responsive behavior, content rules, and tests.

Do not copy vendor components into pages and restyle them independently.

25. Governance

- The frontend and product experience owner owns the design system.
- Safety semantics and shared data meaning require review with the relevant contract owner.
- New tokens require a repeated need across at least two components or screens.
- Page-specific hex colors, radii, spacing values, and status labels are prohibited.
- Breaking changes increment the version and include migration notes.
- Deprecated tokens remain aliases for one migration release.
- Every release records added, changed, deprecated, and removed items.
- Exceptions name the reason, owner, and removal condition.

26. Verification checklist

Before merging UI work:

- The primary question is clear within two seconds.
- Risk is separate from device health, confidence, freshness, and workflow.
- Missing data never appears normal.
- Every used color pair passes contrast checks.
- Keyboard flow completes the main action.
- Focus is visible and restored after overlays.
- Loading, empty, error, stale, disabled, and success states are tested.
- Long and translated text wraps safely.
- Desktop, tablet, narrow, 200% zoom, and coarse-pointer layouts are inspected.
- Grayscale preserves priority.
- Reduced motion and forced colors remain usable.
- No page-local design values were introduced.

27. Research basis

The system uses principles from current primary documentation, not copied screenshots:

- [Inter typeface](#)
- [WCAG 2.2](#)
- [Radix color roles](#)
- [Radix accessibility](#)
- [Carbon data table](#)
- [Adobe Spectrum table](#)
- [USWDS table](#)
- [GOV.UK notification banner](#)

28. Final rule

The new UI should look quiet at rest and unmistakable when action is required. If a styling decision does not help staff understand priority, truth, ownership, or the next action, remove it.